

CRITICAL REVIEW AND INVITED COMMENTARY

Nonconvulsive status epilepticus and coma

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SUMMARY

Nonconvulsive status epilepticus (NCSE) in a comatose patient cannot be diagnosed without electroencephalography (EEG). In many advanced coma stages, the EEG exhibits continuous or periodic EEG abnormalities, but their causal role in coma remains unclear in many cases. To date there is no consensus on whether to treat NCSE in a comatose patient in order to improve the outcome or to retract from treatment, as these EEG patterns might reflect the end stages of a dying brain. On the basis of EEG, NCSE in comatose patients may be classified as generalized or lateralized. This review aims to summarize the ongoing debate of NCSE and coma and to critically reassess the available literature on coma with epileptiform EEG pattern and its prognostic and therapeutic implications. The authors suggest distinguishing NCSE *proper* and comatose NCSE, which includes coma with continuous lateralized

discharges or generalized epileptiform discharges (coma-LED, coma-GED). Although NCSE *proper* is accompanied by clinical symptoms suggestive of status epilepticus and mild impairment of consciousness, such as in absence status or complex focal status epilepticus, coma-LED and coma-GED represent deep coma of various etiology without any clinical motor signs of status epilepticus but with characteristic epileptiform EEG pattern. Hence coma-LED and coma-GED can be diagnosed with EEG only. Subtle or stuporous status epilepticus and *epilepsia partialis continua*-like symptoms in severe acute central nervous system (CNS) disorders represent the borderland in this biologic continuum between NCSE *proper* and comatose NCSE (coma-LED/GED). This pragmatic differentiation could act as a starting point to solve terminologic and factual confusion.

KEY WORDS: Coma, Nonconvulsive status epilepticus, Encephalopathies, EEG, Prognosis.

DEFINITION OF TERMS

Nonconvulsive status epilepticus (NCSE) and coma are subject to divergent definitions. For the purposes of this review we use the following definitions: Loss of consciousness (LOC) is defined operationally when no or inadequate reaction to external stimuli can be obtained and the patient is amnesic for this episode. Coma can be defined as unarousable psychologic unresponsiveness in which the subject lies with eyes closed (Plum & Posner, 1980). The diagnosis is based on a neurologic syndrome encompassing LOC and signs of disturbed brainstem func-

tions (in cases of transtentorial herniations or subtentorial lesions) or *impaired* brain functions down to the midbrain level (e.g., in diffuse encephalopathies) (for summary see Bauer, 2005). Although LOC represents a prerequisite for coma, these terms are not interchangeable. Furthermore, coma has to be differentiated from confusional states, with or without mild impairment of consciousness, (Kaplan, 1999), which are frequently encountered with several types of status epilepticus (SE) and also in the development of coma (Plum & Posner, 1980).

Impairment or LOC, but not coma, can be observed with focal complex and absence seizures and SE of these seizure types. Furthermore, LOC down to the level of coma is transiently present in the ictal and postictal phase of generalized tonic-clonic seizures (GTCS), irrespective of whether primary or secondary generalized. In generalized convulsive SE the postictal coma after a GTCS is followed by another GTCS merging into an epileptic continuum, with progressive electromechanical dissociation leading

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to the advanced stages of stuporous (Clark & Prout, 1903a, 1903b, 1904) or subtle SE (Treiman et al., 1984). These conditions of advanced GCSE can also be termed NCSE, which in fact has been a widely used clinical concept (Engel, 2006). However, the definition of NCSE is not straightforward. The current International League of Epilepsy (ILAE) classifications of seizures and epileptic syndromes (Commission on Classification and Terminology of the ILAE, 1981, 1989) do not encompass SE. The proposed diagnostic scheme for people with epileptic seizures and with epilepsy (Engel, 2001, 2006) includes a section with “*continuous seizure types*” but avoids a definition of SE. Furthermore, the terms “*convulsion*” or “*convulsive*” are omitted as nonspecific lay expressions (Engel, 2001). Nevertheless, NCSE is diagnosed worldwide and denotes an enduring epileptic condition with reduced or altered consciousness, behavioral and vegetative abnormalities, or merely subjective symptoms like auras, but without major convulsive movements (Drislane, 2000). This pragmatic definition implicitly includes epileptiform electroencephalography (EEG) activity. Because a detailed diagnosis according to the underlying seizure types and epilepsy syndrome or neurologic disease is frequently not possible at the time of admission, we fully accept that a generic term such as NCSE is needed. The inclusion of deep coma stages with epileptiform EEG discharges has led to a major conceptional enlargement of the generic term NCSE, with several highly controversial consequences (Kaplan, 2003). Claassen et al. (2004) use the terms “*nonconvulsive*” and “*subclinical*” interchangeably and call the corresponding EEG abnormalities “*electrographic seizures*.” Other terms are *SE in comatose patients*, *generalized electrographic SE* and *non-tonic-clonic SE* (for summary see Brenner, 2002). Regardless of the terminology, comatose NCSE is not listed in the report of the core group (Engel, 2006). In this review we use comatose NCSE to designate a condition with coma accompanied by continuous or periodic epileptiform discharges with or without minor motor activity.

CONCEPTIONAL OVERVIEW AND CLASSIFICATION OF COMATOSE NCSE

From a clinical standpoint, any attempt to classify comatose NCSE has to answer four critical questions: (1) Is the coma caused by an epileptic seizure or SE, or by the underlying brain disorder itself? (2) To what degree does the epileptic activity contribute to the depth of coma? (3) Does the ongoing epileptic activity worsen the prognosis? (4) Should we treat all forms of epileptic activity found in coma?

The epileptic etiology of some comatose states can be suspected by history, the temporal pattern of coma, and

neurologic signs. Most often, the epileptic etiology is exclusively suggested by the presence of continuous epileptiform discharges (Lowenstein & Aminoff, 1992; Jordan, 1999; Brenner, 2005), despite the fact that there are no valid criteria to differentiate the EEG patterns, which are directly related to status from other epiphenomenal electrical abnormalities caused by the underlying brain dysfunction (Brenner, 2004). Regardless of the etiology and pathophysiology, advanced coma stages are frequently accompanied by continuous epileptiform or periodic abnormalities (for summary see Bauer, 2005; Kaplan, 2006). Some authors suggest that NCSE may be regarded as proven, if both the EEG and the clinical state resolve with vigorous antiepileptic drug (AED) treatment (Kaplan, 2002). The reverse does not exclude an epileptic condition; otherwise, all refractory SE would have to be considered as nonepileptic, which is biologically not plausible.

Some authors consider severe metabolic or anoxic encephalopathies with continuous epileptiform EEG abnormalities (Jordan, 1999) and encephalopathies with triphasic waves (Assal et al., 2000) not as SE, whereas others stress their epiphenomenal character (Kaplan, 1996; Young et al., 1996; Aminoff, 1998). The continuous epileptiform pattern found in advanced coma stages might represent an endstage of irreversible coma, which lasts until the EEG becomes *isoelectric*, and the patients *becomes brain dead*. Drislane and Schomer (1994b) considered electrical status in deep coma the result of a severe encephalopathy and the cause of coma, and Kaplan (2002) does not consider coma a result of NCSE, if an acute brain insult itself may be sufficient to cause the condition. To interpret coma in this way seems to be extremely difficult with sufficient accuracy in a given case, especially in the absence of overt structural lesions. However, most cases of comatose NCSE are acute symptomatic with a severe underlying acute brain disorder, and not idiopathic (DeLorenzo et al., 1998; Hesdorffer et al., 1998; Coyetaux et al., 2000; Towne et al., 2000). Furthermore, an additive negative impact of ongoing electroencephalographic discharges to the underlying lesion (Waterhouse et al., 1998), irrespective of whether there is a consensus to call this condition comatose NCSE, cannot be excluded. To date there are no studies investigating the outcomes in patients with comatose NCSE of varied etiology who were treated with AEDs compared to those without such treatment.

Rossetti et al. (2007) use the term *subtle status* synonymously for *NCSE in coma*. The term *subtle epileptic status* was coined to designate the ominous endpoint of difficult-to-treat or advanced untreated convulsive SE (Treiman et al., 1984, 1998). This condition was already described by Clark and Prout (1903a,b, 1904) at the beginning of the 20th century and termed “*stuporous status*.” Continuous spiking or NCSE after termination of generalized

convulsive SE (Jaitly et al., 1997; DeLorenzo et al., 1998) is identical with subtle or stuporous status and its poorer prognosis (Treiman et al., 1998). Privitera et al. (1994) coin this condition *non-tonic-clonic status epilepticus* and point out that it occurs most commonly in the setting of a diffuse brain injury rather than evolving from convulsive SE. In chronic epilepsies, NCSE may follow the spontaneous termination of a GTCS, which should be differentiated from subtle status because of its benign prognosis (Bauer et al., 1982; Fagan & Lee, 1990). These patients are not in coma; their consciousness may be mildly impaired and confusion is prominent. NCSE in these patients can be classified as absence status, atypical absence status, late absence status de novo, or focal complex status according to the underlying epilepsy syndrome. In this review, we define subtle SE as an advanced stage of difficult-to-treat convulsive SE (Treiman et al., 1984).

A frequently used, although not universally accepted, classification of SE is based on two dichotomies: convulsive versus nonconvulsive and generalized versus focal or lateralized (Brenner, 2004). In comatose patients, the EEG is necessary to differentiate between generalized and lateralized NCSE (Brenner, 2005). It has to be stressed that the EEG exhibits transitions of different patterns as well as of generalized and focal or lateralized abnormalities, which can be documented with repeated EEGs or continuous EEG monitoring. These transitions mirror the complex nature and temporal dynamics of the epileptic activity in status. Therefore, the EEG patterns found in comatose NCSE are a shaky ground for unvaried classifications. Disease history, neurologic lateralizing and localizing signs, as well as neuroimaging findings have to be added. Considering the uncertainties to answer the aforementioned conceptional questions, we would like to replace the term generalized or lateralized comatose NCSE with *coma with generalized epileptiform discharges (coma-GED)* and *coma with lateralized epileptiform discharges (coma-LED)*. Hence the term comatose NCSE should be restricted to coma with subtle clinical seizure activity (such as in subtle SE). Etiologic factors and EEG patterns found in coma-GED and coma-LED are given in Table 1.

COMA WITH GENERALIZED EPILEPTIFORM DISCHARGES (COMA-GED)

A number of advanced diffuse brain disturbances have been associated with coma-GED. The most frequently encountered etiology is cerebral hypoxia-ischemia after cardiorespiratory arrest (CRA) (Lowenstein & Aminoff, 1992; Drislane & Schomer, 1994b; Treiman, 1995; Young et al., 1996). The accompanying generalized periodic epileptiform discharges (GPEDs) prompted the designation

Table 1. Etiologic factors and EEG pattern in generalized and lateralized comatose NCSE

	Etiology	EEG pattern
coma-GED	Diffuse primary or secondary brain disturbances (anoxic, toxic, metabolic, infectious, degenerative) Space-occupying lesions with brainstem compression (direct or due to tentorial herniation) ^a Known epilepsies?	Continuous generalized spiking Periodic spiking Burst suppression pattern in different variations Other generalized periodic abnormalities Bilateral triphasic waves
coma-LED	Focal brain lesions (in most cases acutely acquired) In rare cases diffuse abnormalities (aminophylline intoxication, some forms of diabetic coma) Known epilepsies?	Continuous focal spiking PLEDs Bi-PLEDS Unilateral burst suppression pattern Unilateral triphasic waves
^a Might also present as coma-LED. Bi-PLED, bilateral periodic epileptiform discharges; GEDs, generalized epileptiform discharges; LED, lateralized epileptiform discharges; PLED, periodic epileptiform discharges.		

as NCSE (Drislane & Schomer, 1994b). In anoxic coma, myoclonias and a number of other motor abnormalities might be observed (for review see Aichner & Bauer, 2005). These motor signs can be very subtle and disappear with anesthesia. Furthermore, without sedation, deep coma may be accompanied by decorticate or decerebrate posturing, which have to be clearly separated from epileptic seizures. These patients have the worst prognosis, with death in most cases and persistent vegetative state in few survivors. Coma after intoxications with many different substances can exhibit similar GPEDs (for review see Bauer & Bauer, 2005) and was, therefore, also classified as NCSE. The same is true for metabolic (Lowenstein & Aminoff, 1992; Eleftheriadis et al., 2003; Kaplan, 2004) or infectious encephalopathies (Lowenstein & Aminoff, 1992; Carrera et al., 2008); for Alzheimer's and Creutzfeldt-Jacob disease (CJD) (Rees et al., 1999; Armon et al., 2000; Towne et al., 2000; Schwinn et al., 2001; Cohen et al., 2004; Fernández-Torre et al., 2004), for vascular infarcts (Afsar et al., 2003), for patients with cancer (Drislane, 1994; Hormigo et al., 2004; Cavaliere et al., 2006; Blitshteyn & Jaeckle, 2007), and for critically ill patients with various etiologies (Litt et al., 1998; Claassen et al., 2004). In several reports these patients present as a confusional state with various degrees of impaired consciousness down to the level of deep coma. With space-occupying lesions coma arises from traction

or pressure upon the brainstem, whether directly as in subtentorial lesions or via transtentorial herniation as in supratentorial lesions. Advanced and mostly irreversible cases are also characterized by GPEDs (Vespa et al., 1999, 2003; Bauer, 2005).

EEG patterns

A terminology of continuous, rhythmic, or periodic epileptiform EEG patterns in the critically ill patient was proposed by Hirsch et al. (2005). EEG changes in coma-GED consist of continuous generalized spikes and waves (Fig. 1); periodic spikes with flat periods in between (Fig. 2); burst suppression pattern without (Fig. 3) and with spikes (Fig. 4); and varied morphologies, bilateral tripha-

sic waves without spikes (Fig. 5) and with spikes (Fig. 6), and several other generalized periodic or rhythmic abnormalities (for review see Aichner & Bauer, 2005; Bauer, 2005; Husain, 2006). Irrespective of the etiology the given EEG pattern tends to mirror the depth of coma. However, their prognostic implication varies across different etiologies (e.g., intoxication, cerebral hypoxia, brain trauma).

The burst suppression pattern may indicate severe brain dysfunction or a certain level of generalized anesthesia. No valid signs differentiate epileptic burst suppression as it occurs in children with West (Hrachovy & Frost, 2003; Lux, 2007) or Ohtahara syndrome from the patterns found in deep coma and the supposedly nonepileptic patterns found in generalized anesthesia. Because treatment of SE

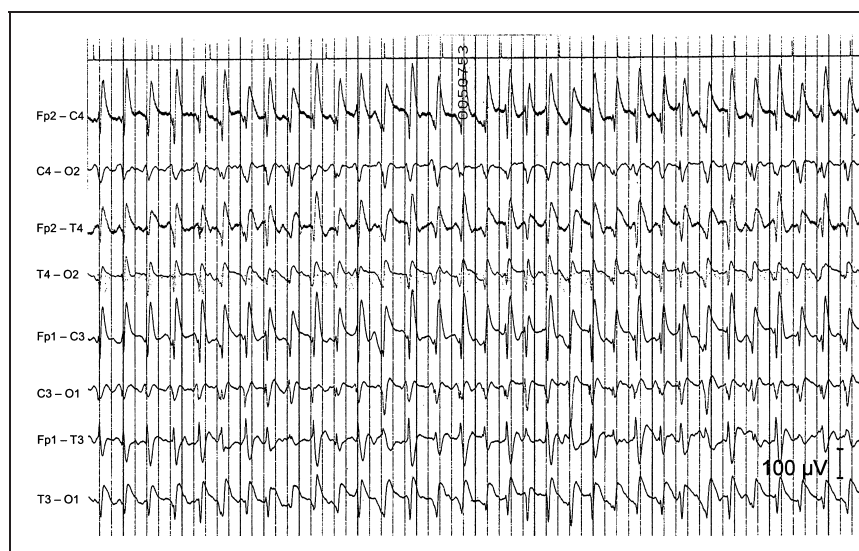


Figure 1.

R.M., female, 76 years.
tc 0.3 HF 30. Comatose after status asthmaticus. Continuous very regular generalized 2–3/s spike and wave activities 8 h after status asthmaticus. No reaction to exogenous stimuli; no change with sedation. Died the next week. tc, time constant.

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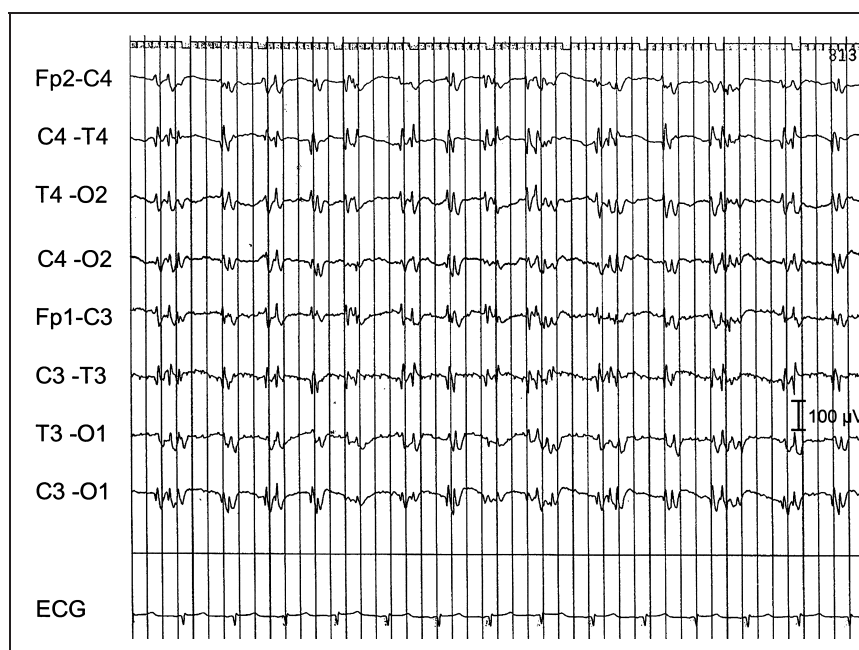


Figure 2.

R.F., male, 63 years.
tc 1.0 HF 30. Coma after cardiac arrest; on respirator. Periodic multiple spikes on a nearly flat background activity on day 2. No response to treatment with antiepileptic drug (AED). Died on day 5 after cardiac arrest.

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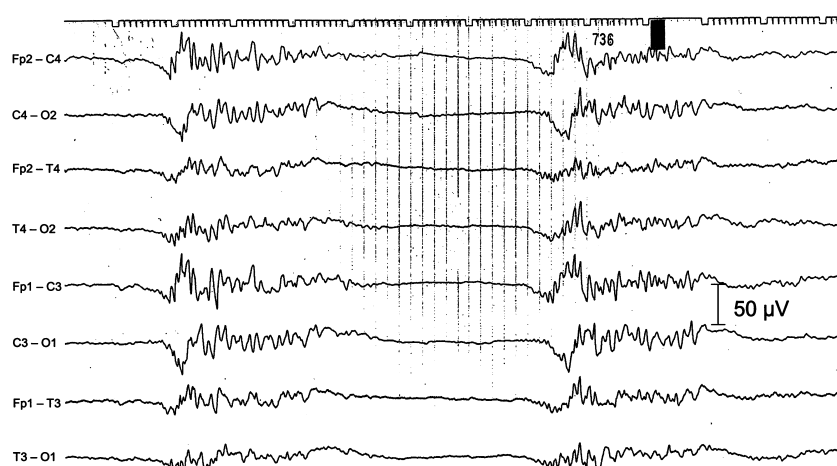


Figure 3.

H.E., female, 49 years. tc 0.3 HF 70. Coma; on respirator. Considered brain dead at neurologic examination. Intoxication with sedative drugs. Suppression burst pattern on the day of admission. Bursts consist of δ -waves superimposed by rhythmic α -frequencies. Long interval between the bursts. She received no antiepileptic drug (AED) treatment. The patient recovered completely.

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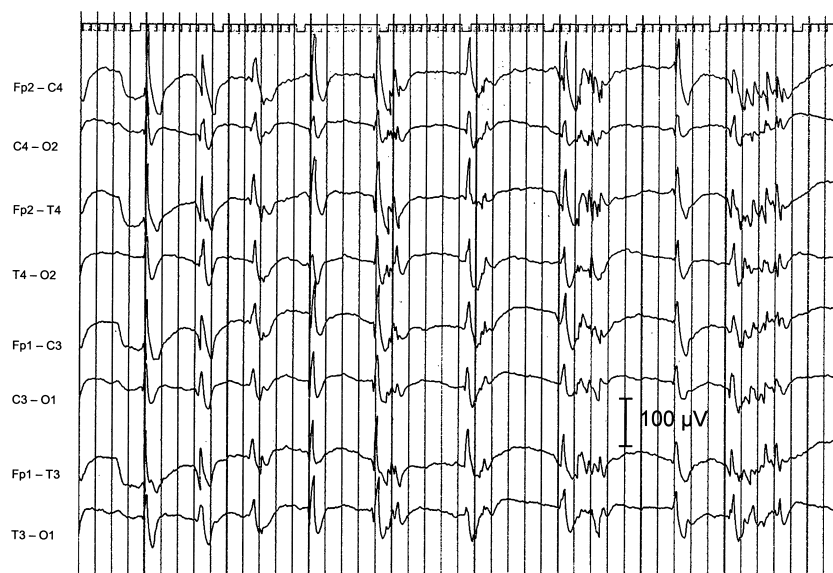


Figure 4.

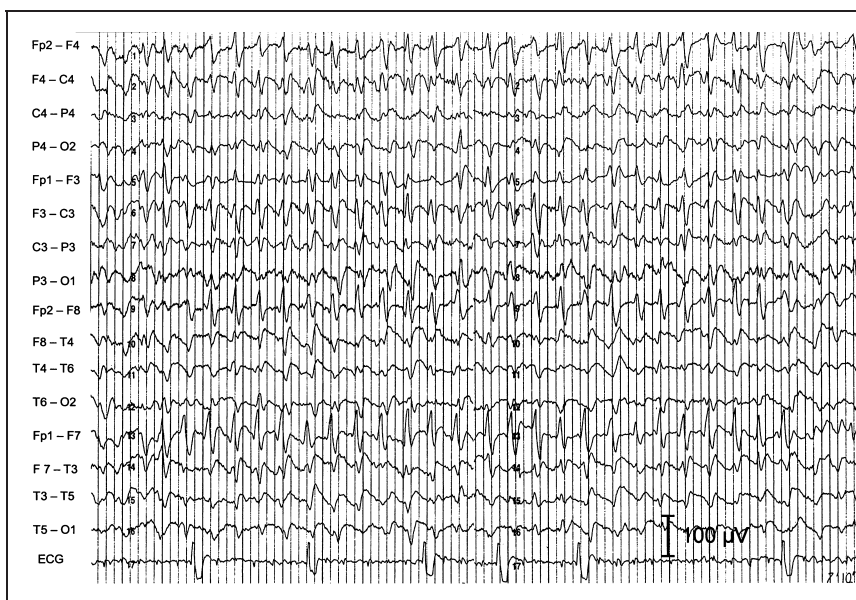
H.W., male, 58 years.
tc 1.0 HF 70. Coma after cardiac arrest; on respirator. Burst suppression pattern containing spikes with short interval on day 3 after the initial event. No response to treatment. Died 2 days later.

Epilepsia © ILAE

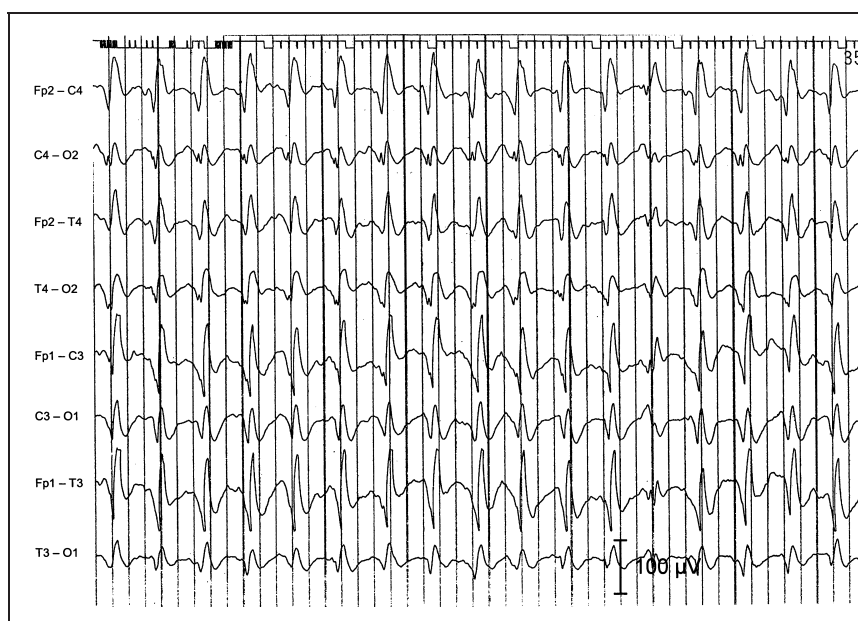
by general anesthesia should be governed by sedation down to burst suppression pattern (Claassen et al., 2001), the question arises of how this can be managed in NCSE characterized by suppression burst pattern itself. The burst suppression duration ratio is claimed to be helpful to determine the end point in treatment of SE (DeGiorgio, 1993), but controlled studies are not available.

Generalized continuously repeated sharp waves or sharp and slow waves with Lennox-Gastaut syndrome are difficult to differentiate from triphasic waves (Blume, 1988), especially in atypical absence status (Fig. 7) (Bauer

et al., 1983). Atypical absence status has been regarded as a possible cause of mental retardation in the Lennox-Gastaut syndrome (Hoffmann-Riem et al., 2000). A clear differentiation of “triphasic-like waves” in NCSE from nonepileptiform “true triphasic waves” in metabolic encephalopathies as suggested by some authors (Husain et al., 1999; Boulanger et al., 2006) would be indeed helpful, but is not yet validated in an observer-blinded study. However, in both instances, the impairment of consciousness is only mild to moderate and almost never reaches the level of coma. In contrast, after cardiopulmonary arrest

**Figure 5.**

N.A.-M., female, 65 years.
tc 1.0 HF 70. Unreactive due to
an advanced stage of Jakob-
Creutzfeldt disease. The record
was taken after 4 mg
intravenous clonazepam. No
change of continuously
repeated 1–2/s triphasic waves,
no spikes. Note irregularities in
electrocardiography (ECG).
Epilepsia © ILAE

**Figure 6.**

Sch.L., female, 60 years.
tc 0.3 HF 70. Coma after
cardiac arrest with irregular
myoclonic twittings.
Generalized 2/s triphasic waves,
preceded by small spikes, best
recognized in channel 2 on day
3 after the initial event. No
change with intravenous
diazepam. Patient died on day 8
after the cardiac arrest.
Epilepsia © ILAE

(CPA) patients are often in deep coma, at the time when triphasic-like waves are recorded (Fig. 6). Their epileptic nature may be suspected if they are accompanied by spikes (coma-GED). However, as Boulanger et al. (2006) pointed out, the spike component was absent in 31% of cases in their series. Again, with coma after CPA there are still no valid criteria differentiating epileptic from nonepileptic triphasic waves.

An EEG pattern represents a time-frozen part of a permanently changing river of abnormalities. Therefore, reliable diagnosis and prognosis with EEG—at least to some extent—is only possible with follow-up examinations or continuous monitoring and careful consideration of the clinical context.

COMA WITH LATERALIZED EPILEPTIFORM DISCHARGES (COMA-LED)

Etiologies leading to coma-LED are similar to those causing coma-GED. Both conditions are dynamic and may show some intermediate forms. A comatose state per se can hardly be regarded as a localized cortical disturbance, since it results from widespread bilateral brain dysfunction. To classify coma as result of a focal cortical lesion, continuous unilateral EEG changes, focal or lateralizing neurologic signs, and unilateral structural brain pathology must be present. Coma may also indicate an

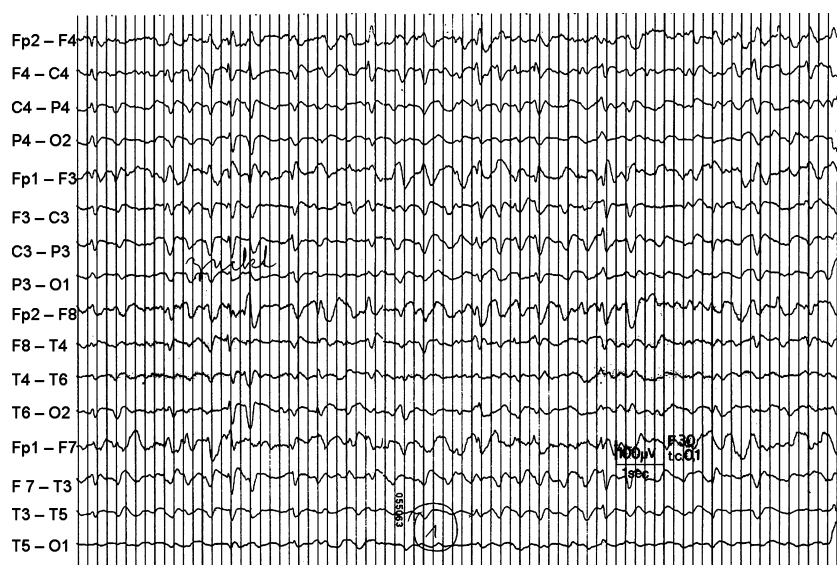


Figure 7.

H.I., female, 33 years to 0.7 HF 30. Late onset Lennox-Gastaut syndrome, seizure onset at age 16 years. Electroencephalography (EEG) taken 3 days after start of an atypical absence status (obtundation, occasional twitches—"zuckt"). Continuously occurring generalized 2/s sharp and slow waves, maximal over the anterior regions. The pattern also can be read as triphasic waves. The EEG and the condition remained unchanged with several intravenous antiepileptic drugs (AEDs) in maximum doses. Spontaneous recovery after 3 weeks.

Epilepsia © ILAE

additional complication of focal or localized central nervous system (CNS) disorders resulting in diffuse or global cerebral dysfunction, like intercurrent hypoxia, due to generalized vasospasm or cardiopulmonary arrest, or increased intracranial pressure with consecutive herniation. Supratentorial space-occupying lesions and other focal lesions like herpes simplex encephalitis or other inflammatory CNS disorders presenting as advanced coma with prominent lateralizing EEG signs are the most frequent etiologies.

EEG patterns

In the center of diagnostic considerations are periodic lateralized epileptiform discharges (PLEDs). A long-lasting debate exists about the assessment of PLEDs as an interictal or ictal EEG pattern. PLEDs occur in the temporal vicinity of overt seizures and reflect an ictal-interictal continuum (Pohlmann-Eden et al., 1996). They can sometimes be observed for weeks and months (chronic PLEDs) in conscious patients with retained alpha or basic rhythm (Westmoreland et al., 1986). Several authors consider comatose or confusional states associated with PLEDs as NCSE (Terzano et al., 1986; Lee & Schauwecker, 1988; Reiher et al., 1992; Handforth et al., 1994; Beaumanoir et al., 1996; Assal et al., 2001; Garzon et al., 2001; Cury et al., 2004; Jette et al., 2006). PLEDs may also occur in fully conscious patients with

epilepsia partialis continua (EPC) (Reiher et al., 1992; Pandian et al., 2004; Striano et al., 2007) as well as negative motor seizures mimicking transitory ischemic attacks (Iriarte et al., 2002). Coma is not a symptom of EPC, but subtle localized EPC-like jerks, which can easily be overlooked, are found in many deeply comatose patients. The EPC-like jerks in these patients constitute a rear-guard action of a full-blown focal motor status. Again the border between coma-GED/LED, comatose NCSE with EPC-like activity, and coma with motor seizures (which is per definition not NCSE) is arbitrary and objective parameters are lacking.

Several types of PLEDs have been distinguished. Reiher et al. (1991) introduced the terms *PLEDs proper* and *PLEDs plus*. The latter consists of PLEDs with intervening rhythmic discharges and superimposed faster frequencies, and is highly associated with seizures. In comatose patients, PLEDs are usually accompanied by diffuse slowing or consist of bilateral (BIPLDs) (Fig. 8) (De La Paz & Brenner, 1981; Nicolai et al., 2001), alternating (Bauer & Niedermeyer, 1979; Bertolucci & Silva, 1992), ipsilateral independent (Silbert et al., 1996), or multifocal PLEDs (Lawn et al., 2000). A longer duration of the interval in between the PLEDs is claimed to point to a cortical origin, in contrast to subcortical PLEDs with a shorter interval (Kalamangalam et al., 2007). PLEDs with flat periods between complexes may be called *unilateral*

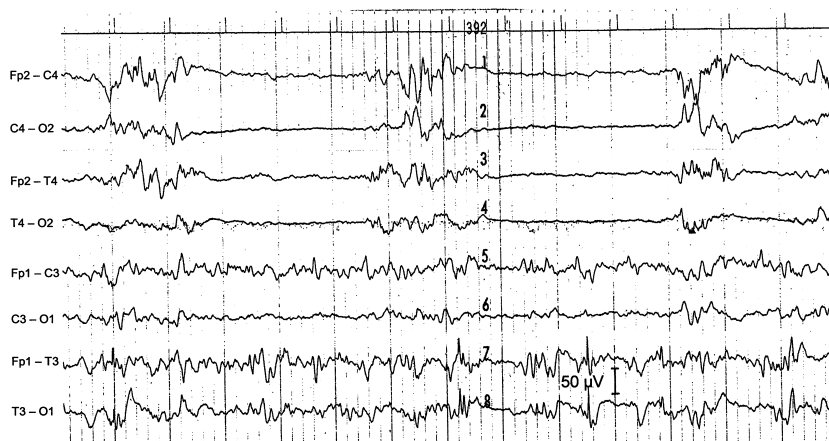


Figure 8.

E.A., male, 53 years. tc 0.3 HF 30. Severe comatose state; on respirator. Had right-sided craniotomy after brain trauma. Bi-PLEDs with unilateral suppression burst pattern over the right hemisphere on day 6 after surgery. EEG was requested due to delayed awakening. Patient survived with spastic tetraparesis more prominent on the left. *Epilepsia* © ILAE

or focal burst suppression (Fig. 8) (Sperling et al., 1986; Hosain et al., 1995). PLEDs over the dominant hemisphere may be associated with aphasia or aphasic status epilepticus (Hirsch et al., 2001; Trinka et al., 2001).

A number of etiologies are found with PLEDs (Chatrian et al., 1964; Bauer et al., 1981; Pohlmann-Eden et al., 1996; García-Morales et al., 2002; Fitzpatrick & Lowry, 2007). The most frequent causes are acute symptomatic seizures with vascular lesions or late seizures after a cerebral infarct. PLEDs can be observed in herpes simplex encephalitis, space-occupying lesions, acute brain trauma, and multiple sclerosis, among others in a long list of focal brain disorders. Continuous EEG monitoring revealed PLEDs in association with electrographic seizures and NCSE as the cause of coma in patients with subarachnoid hemorrhage (Claassen et al., 2005, 2006). PLEDs and seizures can also be recorded with diffuse brain disturbances such as aminophylline intoxication (Yarnell & Nai-Shin-Chu, 1975) or ketotic hyperglycemia (Cokar et al., 2004).

PROGNOSIS AND TREATMENT OF COMA WITH GENERALIZED OR LATERALIZED EPILEPTIFORM DISCHARGES

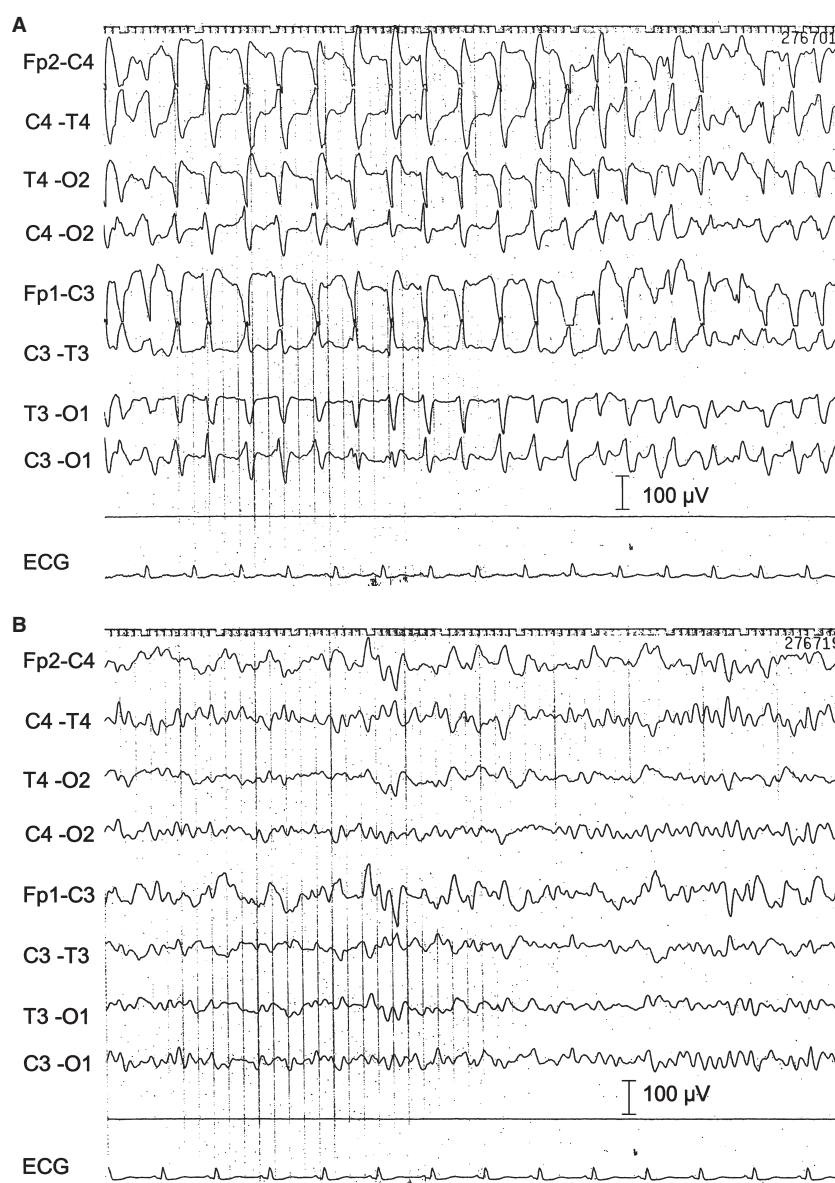
The use of abbreviated EEGs in the emergency room and of serial EEGs or continuous EEG monitoring in intensive care units (ICUs) has increased the number of diagnosed comatose NCSE (Young et al., 1996; Towne et al., 2000; Pandian et al., 2004; Jette et al., 2006; Bautista et al., 2007; Jirsch & Hirsch, 2007; Narayanan &

Murthy, 2007). The prognosis with regard to survival or regaining a meaningful life is dismal in the overwhelming number of cases (Nei et al., 1999; Brenner, 2002). The serious prognosis contrasts treatable NCSE proper with other seizure types (e.g., absence status or complex partial SE). The different prognosis has prompted the concept of “true” or “false” NCSE (Niedermeyer, 2005). However, tractability alone cannot act as a basis for classification, as mentioned in the preceding text. Refractory complex partial SE is also associated with serious morbidity and mortality (Krumholz et al., 1995), and atypical absence status is refractory to AED treatment but resolves without any recognizable cause (Fig. 7) (Bauer et al., 1983).

In comatose NCSE, Waterhouse et al. (1998) suggested a synergistic effect of ischemic brain injury and electrical SE on mortality. Unfortunately, the authors did not report on the influence of AED treatment on prognosis. The question arises of how aggressive AED treatment should be. All patients with coma-GED/LED are treated in ICUs. Patients with respiratory failure and mechanical ventilation are always under generalized anesthesia in ICUs. The anesthetic drugs used in this context have a profound antiepileptic effect per se. Hence none of these patients is left untreated *sensu strictu*. In comatose NCSE and coma-GED/LED with sufficient spontaneous respiration, the decision has to be made whether to add AED treatment—with the possible risk of additional sedation and respiratory depression—to the general intensive care or to take a wait and see position. Pros (Treiman, 1997) and cons (Sharbrough, 1997) have been formulated. Kaplan (1999) pointed out that AED treatment is not without risks for the patient and assumes that NCSE is potentially

Figure 9.

H.F., male, 65 years.
tc 0.3 HF 35. **(A)** Comatose,
unknown etiology, considered
coma after cardiac arrest of
unknown duration. Continuous
2/s triphasic waves on the day
of admission. Note
spontaneous change of
rhythmicity in the second part
of the record. **(B)** Identical
record as with Fig. 9a.
Electroencephalography
activities after termination of
10 mg intravenous diazepam.
Stop of rhythmic triphasic-like
waves, 7/s rhythmical activities
over the posterior regions.
Patient recovered and was
discharged after one week.
Epilepsia © ILAE



overtreated. To date, no formal standard of care can be given on the basis of EEG patterns alone (Chong & Hirsch, 2005). Pragmatically, we would propose a treatment trial with sufficient intravenous doses of an AED after exclusion of intoxications. If the response leads to electrographic and clinical improvement, the AED treatment should be continued (Fig. 9A,B). In a study on comatose NCSE, only 2 of 33 patients improved with AEDs (Drislane & Schomer, 1994a). In our opinion, these rare and surprising positive results justify an *ex juvantibus* trial with intravenous AEDs. However, a cessation of epileptiform EEG abnormalities without any clinical improvement can be observed after application of AEDs, especially in cases with triphasic waves (Fountain & Waldman, 2001). Clearing of EEG activities is transient in many cases and, if not

followed by clinical recovery, there is no reason to continue a potentially harmful sedation. Most EEG abnormalities are not modified at all with AED treatment (e.g., Figs. 5 and 6). In refractory convulsive SE, anesthesia to achieve a burst suppression pattern was recommended (see preceding text of this article). This strategy seems to be questionable in advanced comatose NCSE, especially in cases with burst suppression pattern as a diagnostic finding. Mortality is determined by etiology, severity of coma, and development of acute complications, but not by the type of EEG changes (Shneker & Fountain, 2003). In these patients, the added harm to the brain by continuous epileptiform discharges may be minimal or nothing, and the prognosis purely determined by the underlying disorder. In postanoxic coma, prediction of poor outcome is provided

by combined clinical and electrophysiologic approach, including somatosensory evoked potentials and neuron-specific enolase (Young et al., 2004; Zandbergen et al., 2006). Recovery of EEG activities with early serial EEG can indicate a favorable prognosis (Pressler et al., 2001). Postanoxic SE has been repeatedly associated with poor prognosis (Wijdicks et al., 1994; Rossetti et al., 2007), and practice parameters were developed to assist in predicting poor outcome and in the decision to withdraw therapeutic efforts (Wijdicks et al., 2006). However, some patients with postanoxic SE may have a better prognosis (Arnoldus & Lammers, 1995), which may be even more improved with hypothermia. Rossetti et al. (2009) identified postanoxic SE and preserved brainstem reactions, somatosensory evoked potentials, and EEG reactivity as good prognostic indicators if postanoxic SE was treated vigorously. Although this might support the view that SE adds to the anoxic brain damage, and treatment improves outcome, one could also argue that SE obscures the neurologic picture of the underlying condition, suggesting a poor outcome, but this does not necessarily mean that SE causes additional brain damage.

With intoxications, further sedation due to AEDs should be used cautiously and all attempts to secure the oxygenation of the brain have to be undertaken. The prognosis of coma with GEDs is significantly less serious with intoxications. In these cases coma might be accompanied by otherwise ominous EEG patterns such as burst suppression or triphasic waves (Kubicki, 1967; Kaplan & Birbeck, 2006) (Figs. 4 and 10). The EEG changes and the clinical state may fully resolve to normal if drug intoxications are the principal cause (for review see Bauer & Bauer, 2005; Blume, 2006).

In coma with PLEDs, the prognosis seems to be somewhat less serious. Fitzpatrick and Lowry (2007) found an overall mortality of 27% in 96 consecutive patients with

PLEDs. Mortality among patients with Bi-PLEDs was almost twice as high, with 52%. Unfortunately there was no detailed information on the depth of coma in their series. However, PLEDs can indicate the diagnosis of herpes encephalitis and prompt early antiviral treatment, and in patients with PLEDs and space-occupying lesions a surgical approach has to be considered. Most authors recommend treatment of accompanying motor seizures in any case vigorously. These seizures may themselves indicate a neurologic deterioration.

CONCLUSION AND PROPOSAL FOR A PRAGMATIC CLASSIFICATION

Coma can be accompanied by continuous generalized or localized epileptic discharges (coma-GED/LED) and may, therefore, be called *comatose NCSE*. In sharp contrast to most other types of NCSE, which we propose to designate as *NCSE proper*, the prognosis is serious and almost completely dependent on the underlying disorder causing the coma (Brenner, 2002). Intoxications are an important exception to this rule. Although some EEG patterns have a prognostic value in coma patients, no pattern is specific for a certain etiology or predictive for the outcome independent from the underlying disease. Moreover, aggressive intravenous AED treatment does not show a consistent response in many cases. Studies to investigate the impact of treatment of electroencephalographic SE pattern versus withholding AED treatment on the outcome are lacking and ethically debatable. The differentiation of treatable comatose NCSE from advanced and irreversible coma-GEDs would be bound on identifying EEG features, which are specific for comatose NCSE (Choi et al., 2007). Whether EEG patterns represent an ictal or interictal phenomenon and whether they are the cause of coma or an epiphenomenon of the underlying etiology remains far

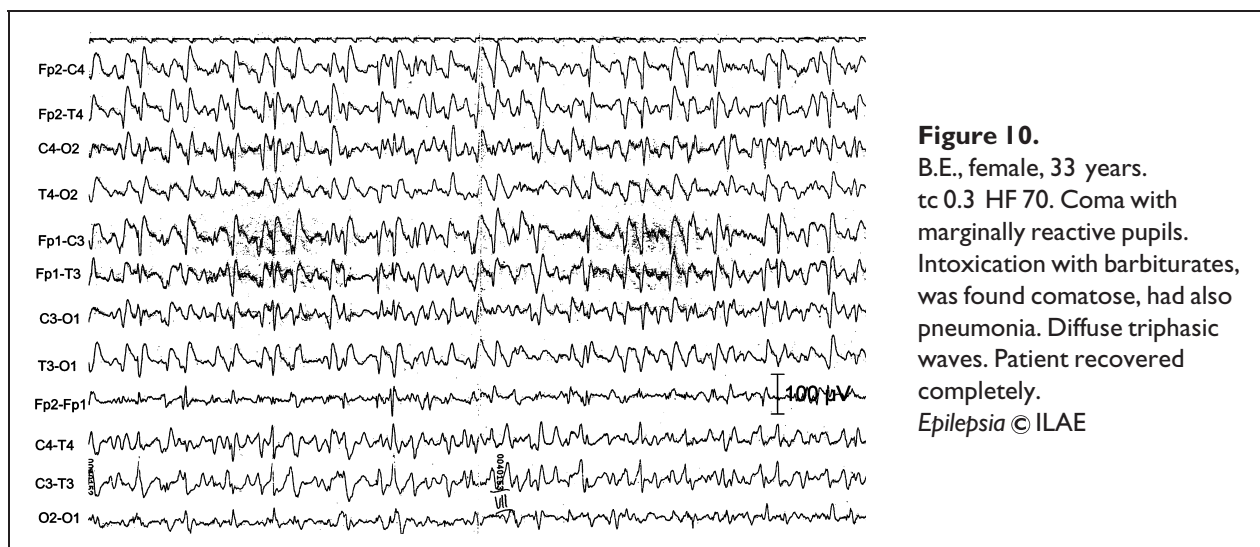


Figure 10.
B.E., female, 33 years.
tc 0.3 HF 70. Coma with
marginally reactive pupils.
Intoxication with barbiturates,
was found comatose, had also
pneumonia. Diffuse triphasic
waves. Patient recovered
completely.
Epilepsia © ILAE

from understood. However, the cellular basic mechanisms might be epileptic in both comatose NCSE and coma-GED/LED.

Once the operational definition of NCSE based on the clinical picture and epileptiform discharges on EEG is accepted, one has to solve the problem of lumping obviously benign conditions, like absence status together with almost always fatal conditions in comatose NCSE, such as postanoxic coma with GED, together under a single term. In order to avoid further confusions it is high time to propose a classification of NCSE that subsumes depth of coma and etiology (Kaplan, 1999). More important than semiologic or linguistic sophistication is to meet the needs of neurology, neurosurgery, pediatric, internal, intensive care and emergency medicine and to use an understandable terminology. We see the various forms of NCSE along a *three-dimensional biological continuum* (Fig. 11). One axis represent the gradual impairment of consciousness and the depth of coma with mild confusion on one end and minimal conscious state just before brain death on the other end. The clinical examination and the EEG pattern correlate loosely along this axis. Another axis represents the degree of structural brain damage, with no magnetic resonance imaging (MRI) visible damage on

one end (e.g., absence status) and severe irreversible cortical damage, such as in coma-GED, on the other end. This axis is largely determining prognosis. The third axis represents the degree to which the ongoing epileptic activity contributes to the depth of coma and the additive structural brain damage due to excessive epileptic activity. The larger the contribution of this axis, the more successful and necessary is vigorous treatment with AEDs. With these dimensions kept in mind one may well accept the clinical category comatose NCSE as a broad range of conditions, with coma-LED and coma-GED as its most extreme forms. Similarly the most extreme forms represent the borderland between an irreversible structural damage in a dying brain and brain death.

Future discussion may come to the consensus to exclude coma-GED/LED from comatose NCSE. This can only been achieved if criteria are developed to reliably distinguish the treatable forms of NCSE from the irreversible forms. As there are no studies aiming to clarify this issue until today and future studies will be ethically questionable we should pragmatically accept the term comatose NCSE with its loose-end coma-GED/LED. Within such a pragmatic classification there is hardly a place for a diagnosis exclusively based on EEG, without considering the etiology and the degree of structural brain damage as well as the depth of coma into consideration.

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We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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REFERENCES

- Afsar N, Kaya D, Aktan S, Aykut-Bingol C. (2003) Stroke and status epilepticus: stroke type, type of status epilepticus, and prognosis. *Seizure* 12:23–27.
- Aichner F, Bauer G. (2005) Cerebral anoxia. In Niedermeyer E, Lopes da Silva F (Eds) *Electroencephalography. Basic principles, clinical applications, and related fields*. 5th ed. Lippincott Williams & Wilkins, Philadelphia, pp. 455–470.
- Aminoff JM. (1998) Do nonconvulsive seizures damage the brain? – No. *Arch Neurol* 55:119–120.
- Armon C, Peterson GW, Liwnicz BH. (2000) Alzheimer's disease underlies some cases of complex partial status epilepticus. *J Clin Neurophysiol* 17:511–518.
- Arnoldus EP, Lammers GJ. (1995) Postanoxic coma: good recovery despite myoclonic status. *Ann Neurol* 38:697–698.
- Assal F, Coeytaux A, Jallon P. (2000) L'état de mal résistant aux antiépileptiques. *Neurophysiol Clin* 30:139–145.
- Assal F, Papazyan JP, Slosman DO, Jallon P, Goerres GW. (2001) SPECT in periodic lateralized epileptiform discharges (PLEDs): a form of partial status epilepticus? *Seizure* 10:260–264.
- Bauer G, Niedermeyer E. (1979) Acute convulsions. *Clin Electroencephalogr* 10:127–144.
- Bauer G, Aichner F, Hengl W. (1981) Der diagnostische Wert periodischer lateralisierter Komplexe im EEG. *Z Elektroenzephalogr Elektromyogr Verwandte Geb* 12:135–141.

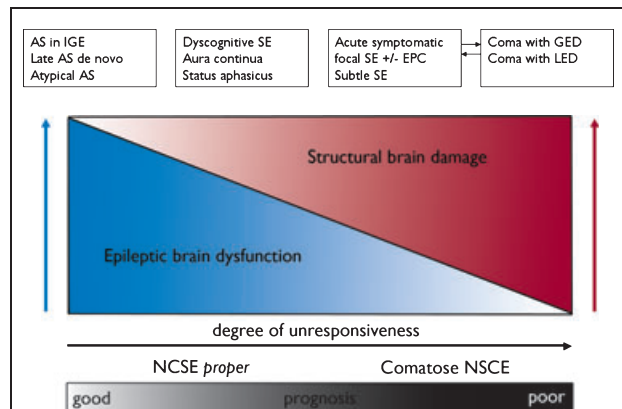


Figure 11.

Relationship between depth of coma (x-axis), prognosis (x-axis), degree of structural brain damage (red y-axis) and epileptic brain dysfunction (blue y-axis) due to status epilepticus. Clinical entities depicted in the upper part of the graph are arranged along the x-axis without distinct positions, in recognition that large border zones and overlaps between the conditions may exist. (For further explanation—see text.) AS, absence status epilepticus, EPC, epilepsia partialis continua, GEDs, generalized epileptiform discharges, IGE, idiopathic generalized epilepsy, LEDs, lateralized epileptiform discharges, NCSE, nonconvulsive status epilepticus.

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- Bauer G, Aichner F, Mayr U. (1982) Nonconvulsive status epilepticus following generalized tonic-clonic seizures. *Eur Neurol* 21:411–419.
- Bauer G, Aichner F, Mayr U. (1983) Status atypischer Absencen im Jugend- und Erwachsenenalter. *Nervenarzt* 54:100–105.
- Bauer G. (2005) Coma and brain death. In Niedermeyer E, Lopes da Silva F (Eds) *Electroencephalography. Basic principles, clinical applications, and related fields*. 5th ed. Lippincott Williams & Wilkins, Philadelphia, pp. 471–487.
- Bauer G, Bauer R. (2005) EEG, drug effect, and central nervous system poisoning. In Niedermeyer E, Lopes da Silva F (Eds) *Electroencephalography. Basic principles, clinical applications, and related fields*. 5th ed. Lippincott Williams & Wilkins, Philadelphia, pp. 701–723.
- Bautista RED, Godwin S, Caro D. (2007) Incorporating abbreviated EEGs in the initial workup of patients who present to the emergency room with mental status changes of unknown etiology. *J Clin Neurophysiol* 24:16–21.
- Beaumanoir A, André-Obadia N, Nahory A, Zerbi D. (1996) Special types of periodic lateralized discharges associated with confusional state in cerebral circulation insufficiency. *Electroencephalogr Clin Neurophysiol* 99:287–292.
- Bertolucci PH, Silva AB. (1992) Alternating periodic lateralized epileptiform discharges (cerebral bigeminy). *Clin Electroencephalogr* 23:177–179.
- Blitshteyn S, Jaecle KA. (2007) Status epilepticus in patients with CNS metastases. *Arch Neurol* 64:916.
- Blume WT. (1988) The EEG features of the Lennox-Gastaut syndrome. In Niedermeyer E, Degen R (Eds) *The Lennox-Gastaut syndrome*, Alan R. Liss, New York, pp. 159–176.
- Blume WT. (2006) Drug effects on EEG. *J Clin Neurophysiol* 23:306–311.
- Boulanger J-M, Deacon C, Lécuyer D, Gosselin S, Reiher J. (2006) Triphasic waves versus nonconvulsive status epilepticus: EEG distinction. *Can J Neurol Sci* 33:175–180.
- Brenner RP. (2002) Is it status? *Epilepsia* 43(suppl 3):103–113.
- Brenner RP. (2004) EEG in convulsive and nonconvulsive status epilepticus. *J Clin Neurophysiol* 21:319–331.
- Brenner RP. (2005) The interpretation of the EEG in stupor and coma. *Neurologist* 11:271–284.
- Carrera E, Claassen J, Oddo M, Emerson RG, Mayer SA, Hirsch LJ. (2008) Continuous Electroencephalographic Monitoring in Critically Ill Patients With Central Nervous System Infections. *Arch Neurol* 65:1612–1618.
- Cavaliere R, Farace E, Schiff D. (2006) Clinical implications of status epilepticus in patients with neoplasms. *Arch Neurol* 63:1746–1749.
- Chatrjian GE, Cheng-Mei Shaw, Leffman H. (1964) The significance of periodic lateralized epileptiform discharges in EEG: an electroencephalographic, clinical and pathological study. *Electroencephalogr Clin Neurophysiol* 17:177–193.
- Choi JS, Krauss G, Kaplan P. (2007) Identifying EEG features of nonconvulsive status epilepticus (NCSE). *Clin Neurophysiol* 118:e185–e186.
- Chong DJ, Hirsch LJ. (2005) Which EEG patterns warrant treatment in the critically ill? Reviewing the evidence for treatment of periodic epileptiform discharges and related patterns. *J Clin Neurophysiol* 22:79–91.
- Claassen J, Hirsch LJ, Emerson RG, Bates JE, Thompson TB, Mayer SA. (2001) Continuous EEG monitoring and midazolam infusion for refractory nonconvulsive status epilepticus. *Neurology* 57:1036–1042.
- Claassen J, Mayer SA, Kowalski RG, Emerson RG, Hirsch LJ. (2004) Detection of electrographic seizures with continuous EEG monitoring in critically ill patients. *Neurology* 62:1743–1748.
- Claassen J, Mayer SA, Hirsch LJ. (2005) Continuous EEG monitoring in patients with subarachnoid hemorrhage. *J Clin Neurophysiol* 22:92–98.
- Claassen J, Hirsch LJ, Frontera JA, Fernandez A, Schmidt M, Kapinos G, Wittman J, Connolly ES, Emerson RG, Mayer SA. (2006) Prognostic significance of continuous EEG monitoring in patients with poor grade subarachnoid hemorrhage. *Neurocrit Care* 4:103–112.
- Clark LP, Prout TP. (1903a) Status epilepticus: a clinical and pathological study (part 1). *Am J Insanity* 60:291–306.
- Clark LP, Prout TP. (1903b) Status epilepticus: a clinical and pathological study (part 2). *Am J Insanity* 60:645–675.
- Clark LP, Prout TP. (1904) Status epilepticus: a clinical and pathological study (part 3). *Am J Insanity* 61:81–108.
- Cohen D, Kutluay E, Edwards J, Peltier A, Beydoun A. (2004) Sporadic Creutzfeldt-Jakob disease presenting with nonconvulsive status epilepticus. *Epilepsy Behav* 5:792–796.
- Cokar Ö, Aydin B, Özer F. (2004) Non-ketotic hyperglycaemia presenting as epilepsy partialis continua. *Seizure* 13:264–269.
- Commission on Classification and Terminology of the ILAE (1981) Proposal for revised clinical and electroencephalographic classification of epileptic seizures. *Epilepsia* 22:489–501.
- Commission on Classification and Terminology of the ILAE (1989) Proposal for revised classification of epilepsies and epileptic syndromes. *Epilepsia* 30:389–399.
- Coyetaux A, Jallon P, Galobardes B, Morabia A. (2000) Incidence of French speaking Switzerland (EPISTAR). *Neurology* 55:693–697.
- Cury RF, Wichert-Ana L, Sakamoto AC, Fernandes RMF. (2004) Focal nonconvulsive status epilepticus associated to PLEDs and intense focal hyperemia in an AIDS patient. *Seizure* 13:358–361.
- De La Paz D, Brenner RP. (1981) Bilateral independent periodic lateralized epileptiform discharges. Clinical significance. *Arch Neurol* 38:713–715.
- DeGiorgio DM. (1993) Burst-suppression patterns in status epilepticus. The burst-suppression duration ratio: a marker of the end of clinical status epilepticus. *J Epilepsy* 6:32–35.
- DeLorenzo RJ, Waterhouse EJ, Towne AR, Boggs JG, DeLorenzo GA, Brown A, Garnett L. (1998) Persistent nonconvulsive status epilepticus after the control of convulsive status epilepticus. *Epilepsia* 39:833–840.
- Drislane FW. (1994) Nonconvulsive status epilepticus in patients with cancer. *Clin Neurol Neurosurg* 96:314–318.
- Drislane FW, Schomer DL. (1994a) Anticonvulsant medication effect in patients with continuous epileptiform discharges. *Clin Neuropharmacol* 17:165–174.
- Drislane FW, Schomer DL. (1994b) Clinical implications of generalized electroencephalographic status epilepticus. *Epilepsy Res* 19:111–121.
- Drislane FW. (2000) Presentation, evaluation, and treatment of nonconvulsive status epilepticus. *Epilepsy & Behaviour* 1:301–314.
- Eleftheriadis N, Fourla E, Eleftheriadis D, Karlovasitou A. (2003) Status epilepticus as a manifestation of hepatic encephalopathy. *Acta Neurol Scand* 107:142–144.
- Engel J. (2001) A proposed diagnostic scheme for people with epileptic seizures and with epilepsy: report of the ILAE task force on classification and terminology. *Epilepsia* 42:796–803.
- Engel J. (2006) Report of the ILAE classification core group. *Epilepsia* 47:1558–1568.
- Fagan KJ, Lee SI. (1990) Prolonged confusion following convulsions due to generalized nonconvulsive status epilepticus. *Neurology* 40:1689–1694.
- Fernández-Torre JL, Solar DM, Astudillo A, Cereceda R, Acebes A, Calatayud MT. (2004) Creutzfeldt-Jakob disease and non-convulsive status epilepticus: a clinical and electroencephalographic follow-up study. *Clin Neurophysiol* 115:316–319.
- Fitzpatrick W, Lowry N. (2007) PLEDs: clinical correlates. *Can J Neurol Sci* 34:443–450.
- Fountain NB, Waldman WA. (2001) Effects of benzodiazepines on triphasic waves: implications for nonconvulsive status epilepticus. *J Clin Neurophysiol* 18:345–352.
- García-Morales I, García MT, Galán-Dávila L, Gómez-Eccalonilla C, Saiz-Díaz R, Matrínez-Salio A, de la Peña P, Tejerina JA. (2002) Periodic lateralized epileptiform discharges. Etiology, clinical aspects, seizures, and evolution in 130 patients. *J Clin Neurophysiol* 19:172–177.
- Garzon E, Fernandes RMF, Sakamoto AC. (2001) Serial EEG during human status epilepticus. Evidence for PLED as an ictal pattern. *Neurology* 57:1175–1183.
- Handforth A, Jung Tung Cheng, Mandelkern MA, Treiman DM. (1994) Markedly increased mesiotemporal lobe metabolism in a case with PLEDs: further evidence that PLEDs are a manifestation of partial status epilepticus. *Epilepsia* 35:876–881.

- Hesdorffer DC, Logroscino G, Cascino G, Annegers JF, Hauser WA. (1998) Incidence of status epilepticus in Rochester, Minnesota 1965–1984. *Neurology* 50:735–741.
- Hirsch LJ, Emerson RG, Pedley TA. (2001) Prolonged “postictal” aphasia: demonstration of persistent ictal activity with intracranial electrodes. *Neurology* 56:134–136.
- Hirsch LJ, Brenner RP, Drislane FW, So E, Kaplan PW, Jordan KG, Herman ST, LaRoche SM, Young B, Bleck TP, Scheuer ML, Emerson RG. (2005) The ACNS subcommittee on research terminology for continuous EEG monitoring: proposed standardized terminology for rhythmic and periodic EEG patterns encountered in critically ill patients. *J Clin Neurophysiol* 22:128–135.
- Hoffmann-Riem M, Diener W, Benninger C, Rating D, Unnebrink K, Stephani U, Ernst HP, Korinthenberg R. (2000) Nonconvulsive status epilepticus – A possible cause of mental retardation in patients with Lennox-Gastaut syndrome. *Neuropediatrics* 31:169–174.
- Hormigo A, Liberato B, Lis E, DeAngelis LM. (2004) Nonconvulsive status epilepticus in patients with cancer. Imaging abnormalities. *Arch Neurol* 61:362–365.
- Hosain S, Burton L, Fraser R, Labar D. (1995) Focal suppression-burst on electrocorticography after temporal lobectomy. *Neurology* 45:2276–2278.
- Hrachovy RA, Frost JD. (2003) Infantile epileptic encephalopathy with hypsarrhythmia (infantile spasms/West syndrome). *J Clin Neurophysiol* 20:408–425.
- Husain AM. (2006) Electroencephalographic assessment of coma. *J Clin Neurophysiol* 23:208–220.
- Husain AM, Mebust KA, Radtke RA. (1999) Generalized periodic epileptiform discharges: etiologies, relationship to status epilepticus, and prognosis. *J Clin Neurophysiol* 16:51–58.
- Iriarte J, Urrestarazu E, Artieda J, Alegre M, Schlumberger E, Lázaro D, Viteri C. (2002) Ictal paralysis mimicking Todd's phenomenon. *Neurology* 59:464–465.
- Jaitly R, Sgro JA, Towne AR, Ko D, DeLorenzo RJ. (1997) Prognostic value of EEG monitoring after status epilepticus: a prospective adult study. *J Clin Neurophysiol* 14:326–334.
- Jette N, Claassen J, Emerson RG, Hirsch LJ. (2006) Frequency and predictors of nonconvulsive seizures during continuous electroencephalographic monitoring in critically ill children. *Arch Neurol* 63:1750–1755.
- Jirsch J, Hirsch LJ. (2007) Nonconvulsive seizures: developing a rational approach to the diagnosis and management in the critically ill population. *Clin Neurophysiol* 118:1660–1670.
- Jordan KG. (1999) Nonconvulsive status epilepticus in acute brain injury. *J Clin Neurophysiol* 16:332–340.
- Kalamangalam G, Diehl B, Burgess RC. (2007) Neuroimaging and neurophysiology of periodic lateralized epileptiform discharges: observations and hypotheses. *Epilepsia* 48:1396–1405.
- Kaplan PW. (1996) Nonconvulsive status epilepticus in the emergency room. *Epilepsia* 37:643–650.
- Kaplan PW. (1999) Assessing the outcome in patients with nonconvulsive status epilepticus: nonconvulsive status epilepticus is underdiagnosed, potentially overtreated, and confounded by comorbidity. *J Clin Neurophysiol* 16:341–352.
- Kaplan PW. (2002) Behavioral manifestations of nonconvulsive status epilepticus. *Epilepsy Behav* 3:122–139.
- Kaplan PW. (2003) Nonconvulsive status epilepticus. *Neurology* 61:1035–1036.
- Kaplan PW. (2004) The EEG in metabolic encephalopathy and coma. *J Clin Neurophysiol* 21:307–318.
- Kaplan PW. (2006) The EEG of status epilepticus. *J Clin Neurophysiol* 23:221–229.
- Kaplan PW, Birbeck G. (2006) Lithium-induced confusional states: non-convulsive status epilepticus or triphasic encephalopathy? *Epilepsia* 47:2071–2074.
- Krumholz A, Sung GY, Fisher RS, Barry E, Bergey GK, Grattan LM. (1995) Complex partial status epilepticus accompanied by serious morbidity and mortality. *Neurology* 45:1499–1504.
- Kubicki S. (1967) Triphasic potentials following a state of deep coma (coma dépassé) in patients with intoxications by hypnotics. *Electroencephalogr Clin Neurophysiol* 23:382.
- Lawn ND, Westmoreland BF, Sharbrough FW. (2000) Multifocal periodic lateralized epileptiform discharges (PLEDs): EEG features and clinical correlations. *Clin Neurophysiol* 111:2125–2129.
- Lee BI, Schauwecker DS. (1988) Regional cerebral perfusion in PLEDs: a case report. *Epilepsia* 29:607–611.
- Litt B, Wityk RJ, Hertz SH, Mullen PD, Weiss H, Ryan DD, Henry TR. (1998) Nonconvulsive status in the critically ill elderly. *Epilepsia* 39:1194–1202.
- Lowenstein DH, Aminoff MJ. (1992) Clinical, EEG features of status epilepticus in comatose patients. *Neurology* 42:100–104.
- Lux AL. (2007) Is hypsarrhythmia a form of non-convulsive status epilepticus in infants? *Acta Neurol Scand* 115:37–44.
- Narayanan JT, Murthy JMK. (2007) Nonconvulsive status epilepticus in a neurological intensive care unit: profile in a developing country. *Epilepsia* 48:900–906.
- Nei M, Lee J-M, Shanker VL, Sperling MR. (1999) The EEG and the prognosis in status epilepticus. *Epilepsia* 40:157–163.
- Nicolai J, van Putten MJAM, Tavy DLJ. (2001) BIPLEDs in akinetic mutism caused by bilateral anterior cerebral artery infarction. *Clin Neurophysiol* 112:1726–1728.
- Niedermeyer E. (2005) Epileptic seizure disorders. In Niedermeyer E, Lopes da Silva F (Eds) *Electroencephalography. Basic principles, clinical applications, and related fields*. 5th ed. Lippincott Williams & Wilkins, Philadelphia, pp. 505–619.
- Pandian JD, Cascino GD, So EL, Manno E, Fulgham JR. (2004) Digital video-electroencephalographic monitoring in the neurological-neurosurgical intensive care unit. *Arch Neurol* 61:1090–1094.
- Plum F, Posner JB (1980) *The diagnosis of stupor and coma*. 3rd ed. F.A. Davis Company, Philadelphia.
- Pohlmann-Eden B, Hoch DB, Cochius JJ, Chiappa KH. (1996) Periodic lateralized epileptiform discharges – A critical review. *J Clin Neurophysiol* 13:519–530.
- Pressler RM, Boylan GB, Morton M, Binnie CD, Rennie JM. (2001) Early serial EEG in hypoxic ischaemic encephalopathy. *Clin Neurophysiol* 112:31–37.
- Privitera M, Hoffman M, Moore JL, Jester D. (1994) EEG detection of nontonic-clonic status epilepticus in patients with altered consciousness. *Epilepsy Res* 18:155–166.
- Rees JH, Smith SJ, Kullmann DM, Hirsch NP, Howard RS. (1999) Creutzfeldt-Jakob disease presenting as complex partial status epilepticus: a report of two cases. *J Neurol Neurosurg Psychiatry* 66:406–407.
- Reiher J, Rivest J, Grand'Maison F, Leduc CP. (1991) Periodic lateralized epileptiform discharges with transitional rhythmic discharges: association with seizures. *Electroencephalogr Clin Neurophysiol* 71:12–17.
- Reiher J, Grand'Maison F, Leduc CP. (1992) Partial status epilepticus: short-term prediction of seizure outcome from on-line EEG analysis. *Electroencephalogr Clin Neurophysiol* 82:17–22.
- Rossetti AO, Logroscino G, Liaudet L, Ruffieux C, Ribordy V, Schaller MD, Despland A, Oddo M. (2007) Status epilepticus. An independent outcome predictor after cerebral anoxia. *Neurology* 69:255–260.
- Rossetti AO, Oddo M, Liaudet L, Kaplan PW. (2009) Predictors of awakening from postanoxic status epilepticus after therapeutic hypothermia. *Neurology* 72:744–749.
- Schwinn PJ, Krumholz A, Seiden LG. (2001) Jakob-Creutzfeldt disease presenting as non-convulsive status epilepticus. *Epilepsia* 42:S146–S147.
- Sharbrough FW. (1997) Controversies in clinical neurophysiology: Which EEG patterns of status epilepticus warrant emergent pharmacologic treatment. *J Clin Neurophysiol* 14:159.
- Shneker BF, Fountain NB. (2003) Assessment of acute morbidity and mortality in nonconvulsive status epilepticus. *Neurology* 61:1066–1073.
- Silbert PL, Radhakrishnan K, Sharbrough FW, Klass DW. (1996) Ipsilateral independent periodic lateralized epileptiform discharges. *Electroencephalogr Clin Neurophysiol* 98:223–226.
- Sperling MR, Brown WJ, Crandall PH. (1986) Focal burst-suppression induced by Thiopental. *Electroencephalogr Clin Neurophysiol* 63:203–208.

- Striano P, Tortora F, Evoli A, Palmieri G, Elefante A, Zara F, Tarr PE, Striano S. (2007) Periodic myoclonus due to Cytomegalovirus encephalitis in a patient with Good syndrome. *Arch Neurol* 64:277–279.
- Terzano MG, Parrino L, Mazzucchi A, Moretti G. (1986) Confusional states with periodic lateralized epileptiform discharges (PLEDs): a peculiar epileptic syndrome in the elderly. *Epilepsia* 27:446–457.
- Towne AR, Waterhouse EJ, Boggs JG, Garnett LK, Brown AJ, Smith JR, DeLorenzo RJ. (2000) Prevalence of nonconvulsive status epilepticus in comatose patients. *Neurology* 54:340–345.
- Treiman DM. (1995) Electroclinical features of status epilepticus. *J Clin Neurophysiol* 12:343–362.
- Treiman DM. (1997) Controversies in clinical neurophysiology: which EEG patterns of status epilepticus warrant emergent treatment? *J Clin Neurophysiol* 14:159.
- Treiman DM, DeGiorgio CMA, Salisbury SM, Wickboldt CL. (1984) Subtle generalized convulsive status epilepticus. *Epilepsia* 25:653.
- Treiman DM, Meyers PD, Walton NY, Collins JF, Colling C, Rowan AJ, Handforth A, Faught E, Calabrese VP, Uthman BM, Ramsay RE, Mamdani MB. (1998) A comparison of four treatments for generalized convulsive status epilepticus. *N Engl J Med* 339:792–798.
- Trinka E, Unterberger I, Benke T, Berger T, Bauer G. (2001) Acute aphasia in multiple sclerosis. *Arch Neurol* 58:133–134.
- Vespa PM, Nuwer MR, Nenov V, Ronne-Engstrom E, Hovda DA, Bergsneider M, Kelly DF, Martin NA, Becker DP. (1999) Increased incidence and impact of nonconvulsive and convulsive seizures after traumatic brain injury as detected by continuous electroencephalographic monitoring. *J Neurosurg* 91:750–760.
- Vespa PM, O'Phelan K, Shah M, Mirabelli J, Starkman S, Kidwell C, Saver J, Nuwer MR, Frazee JG, McArthur DA, Martin NA. (2003) Acute seizures after intracerebral hemorrhage. A factor in progressive midline shift and outcome. *Neurology* 60:1441–1446.
- Waterhouse EJ, Vaughan JK, Barnes TY, Boggs JG, Towne AR, Kopec-Garnett L, DeLorenzo RJ. (1998) Synergistic effect of status epilepticus and ischemic brain injury on mortality. *Epilepsy Res* 29:175–183.
- Westmoreland BF, Klass DW, Sharbrough FW. (1986) Chronic periodic epileptiform discharges. *Arch Neurol* 6:713–715.
- Wijdicks EFM, Parisi JE, Sharbrough FW. (1994) Prognostic value of myoclonic status epilepticus in comatose survivors of cardiac arrest. *Ann Neurol* 35:239–243.
- Wijdicks EFM, Hijdra A, Young GB, Bassetti CL, Wiebe S. (2006) Practice parameter: prediction of outcome in comatose survivors after cardiopulmonary resuscitation (an evidence-based review). *Neurology* 67:203–210.
- Yarnell PR, Nai-Shin-Chu C. (1975) Focal seizures and aminophylline. *Neurology* 25:819–822.
- Young GB, Jordan KG, Doig GS. (1996) An assessment of nonconvulsive seizures in the intensive care unit using continuous EEG monitoring: an investigation of variables associated with mortality. *Neurology* 47:83–89.
- Young GB, Wang JT, Connolly JF. (2004) Prognostic determination in anoxic-ischemic and traumatic encephalopathies. *J Clin Neurophysiol* 21:379–390.
- Zandbergen EGJ, Hijdra A, Koelman JHTM, Hart AAM, Vos PE, Verbeek MM, de Haan RJ, for the PROPAC Study Group (2006) Prediction of poor outcome within the first 3 days of postanoxic coma. *Neurology* 66:62–68.